

CSE5DMI Data Mining

Week 04
CLASSIFICATION: BASIC CONCEPTS, DECISION TREES, AND MODEL EVALUATION

La Trobe University CRICOS Provider Code Number 00115M

Today's Learning Journey

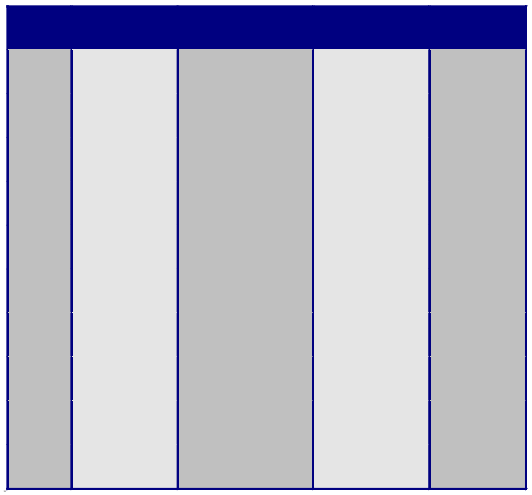
- Understand underfitting and overfitting
- Evaluate how well models perform on unseen data
- Apply train/test splits, cross-validation and confusion matrices

RUNNING EXAMPLE

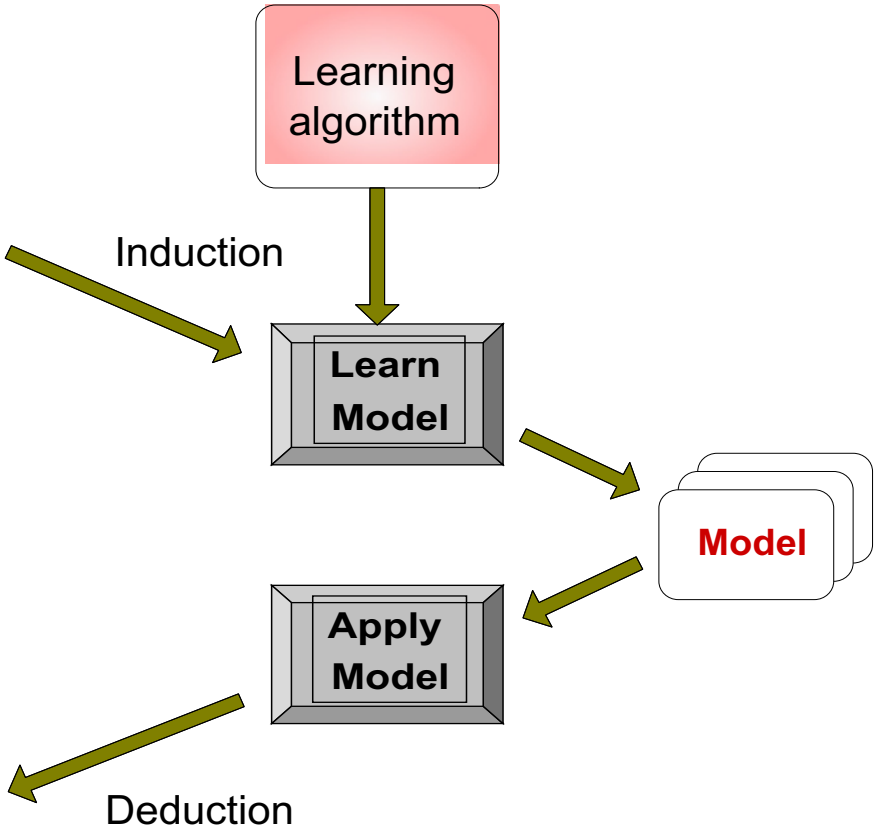
Can a bank's loan model predict correctly for a new customer?

LA TROBE UNIVERSITY
All kinds of clever

Review – Classification Task



Training Set



Test Set

Summary:
Training Set
↓
Learning Algorithm
(e.g. Decision Tree)
↓
Learn (or Build) a Model
↓
Test Set

LA TROBE UNIVERSITY
All kinds of clever

Quick Quiz: Classification

Which item is the model's output?

A. Income

B. Credit score

C. Approved or Rejected

D. Existing debt

Explain your choice.

LA TROBE UNIVERSITY
All kinds of clever

Case Study: A New Applicant – Unseen Data – understanding feature input and output class

A customer applies for a loan:

Income: $62,000
Employment: Full-time
Credit score: 690
Existing debt: $18,000

Question: What information enters the classifier—and what prediction must it produce?

What are feature inputs and output?

LA TROBE UNIVERSITY
All kinds of clever

Answer – Understanding Feature Input and Output

Inputs: income, employment, credit score and existing debt

Output class: Approved or Rejected

Key idea: **Classification** maps input features to a class label.

Next question: A classifier does not always make the correct prediction. How can we measure whether its predictions are accurate or how often it makes errors?

LA TROBE UNIVERSITY
All kinds of clever

Three Types of Error

DEFINITIONS

Training error: mistakes on data used to build the model.

Test error: mistakes on held-out, unseen data.

Generalisation error: The **expected error on future unseen data** from the same population. It **cannot be measured directly**.

LOAN EXAMPLE

Training: 1 mistake from 5 applications = 20%.

Test: 3 mistakes from 5 unseen applications = 60%.

The test result estimates future performance.

Question: Which result should the bank trust more before deployment?

Summary:
Training Error (Measured)

Test Error (Measured)

Generalisation Error (Estimated)

LA TROBE UNIVERSITY
All kinds of clever

Training Error Example

Training error = incorrect training predictions ÷ all training examples.

Customer	Actual Label	Prediction (*Using Traning Data*)	Result
A	Approved	Approved	✅
B	Rejected	Rejected	✅
C	Approved	Rejected	❌
D	Rejected	Rejected	✅
E	Approved	Approved	✅

The model makes **1 mistake out of 5**:

LA TROBE UNIVERSITY
All kinds of clever

Test Error Example

Test error = incorrect test predictions ÷ all test examples.

Customer	Actual	Prediction	Result
F	Approved	Rejected	❌
G	Rejected	Rejected	✅
H	Approved	Rejected	❌
I	Rejected	Approved	❌
J	Approved	Approved	✅

The model makes **3 mistakes out of 5**:

LA TROBE UNIVERSITY
All kinds of clever

Generalisation Error

Future Example(s)

Generalisation error is the **expected error on future unseen data that follow the same distribution as the training data.**

Since we cannot evaluate the model on all future data, the **true generalisation error is unknown**.

Therefore, we use the **test error** measured on a representative unseen test set as our **best estimate of the generalisation error**.

Future unseen customers

↑

Generalisation Error (unknown)

↑ estimated by

Test Error (measured on unseen test data)

The future customers are **not the same people** in our training or test set.

They are **new customers**, but we assume they come from the **same distribution**.

For example, they are applying for loans under similar business conditions, with similar characteristics and class patterns.

LA TROBE UNIVERSITY
All kinds of clever

Three Views of Error

ONE CLASSIFIER

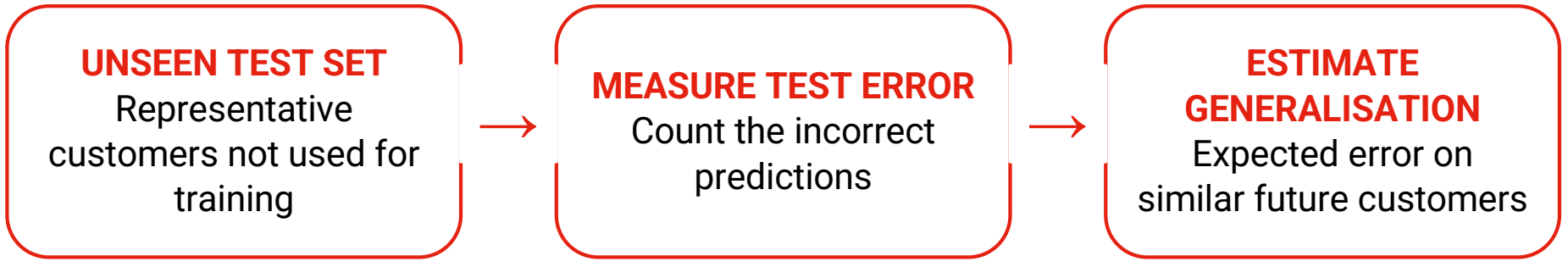
TRAINING SET	TEST SET	FUTURE SIMILAR DATA
Used to build the model	Held out and unseen	Where the model is deployed
Training error: mistakes on familiar data	Test error: mistakes on test data	Generalisation error: **expected future error**

Test error is our practical estimate of generalisation error.

LA TROBE UNIVERSITY
All kinds of clever

Generalisation to New Data

Generalisation means performing well on new, unseen data.



The test set must represent the same population as future data.

LA TROBE UNIVERSITY
All kinds of clever

Why is Generalisation Important? Low Training Error Alone Is Not Enough

- A good classifier should perform well on **both the training data and unseen data**.
- A model that performs well only on the training data may **fail on new unseen data**.
- **Very low training error** but **high test error** is a sign of **overfitting**.

LA TROBE UNIVERSITY
All kinds of clever

Illustration: Creating Model From Five Training Samples

Use these five labelled applicants to learn a simple decision boundary.

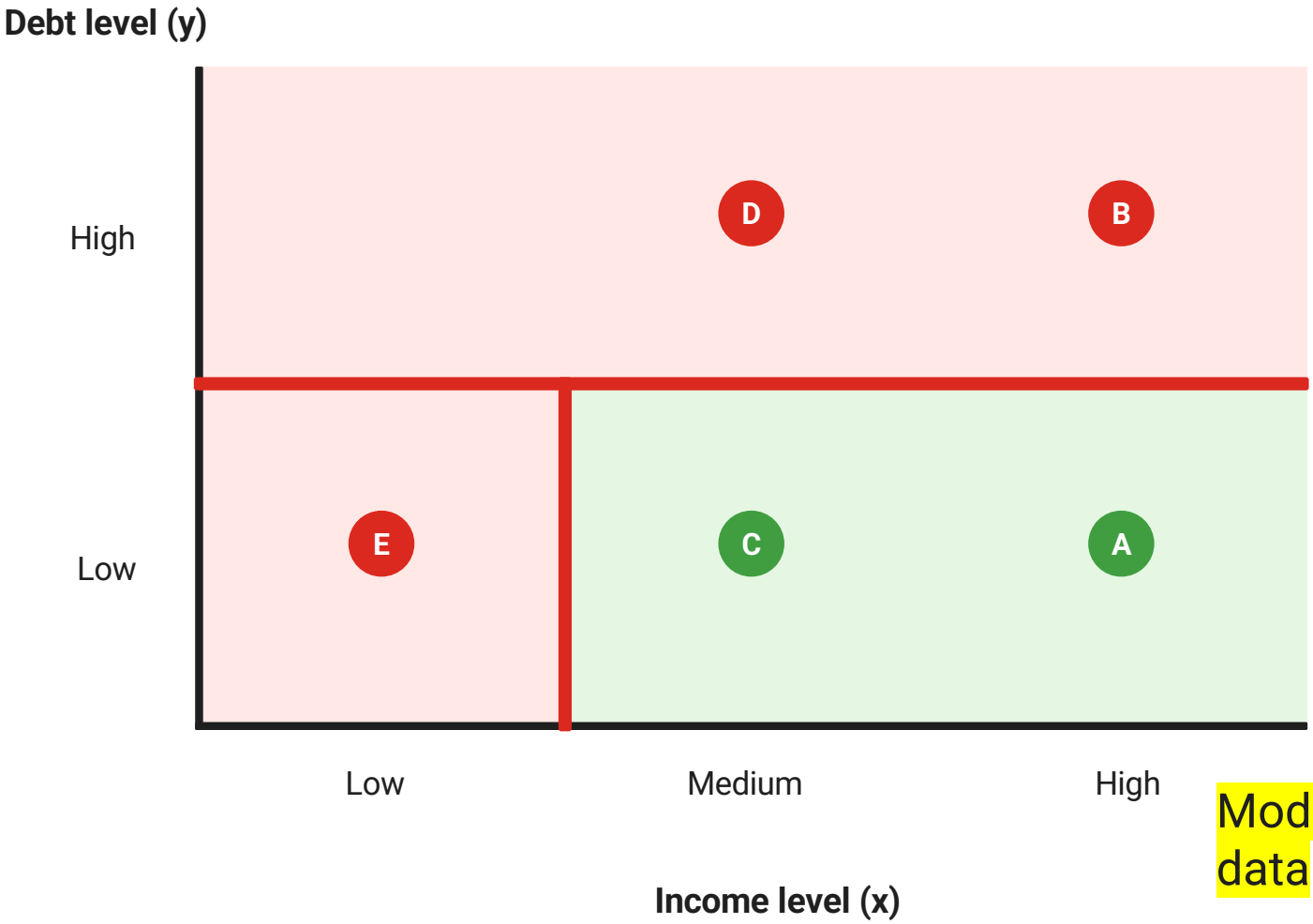
Applicant	Income	Debt	Actual class
A	High	Low	Approved
B	High	High	Rejected
C	Medium	Low	Approved
D	Medium	High	Rejected
E	Low	Low	Rejected

Observation: the learned boundary classifies all 5 training samples correctly.

LA TROBE UNIVERSITY
All kinds of clever

Training Boundary: 5/5 Correct

The model learns one fixed rule from the five training applicants.



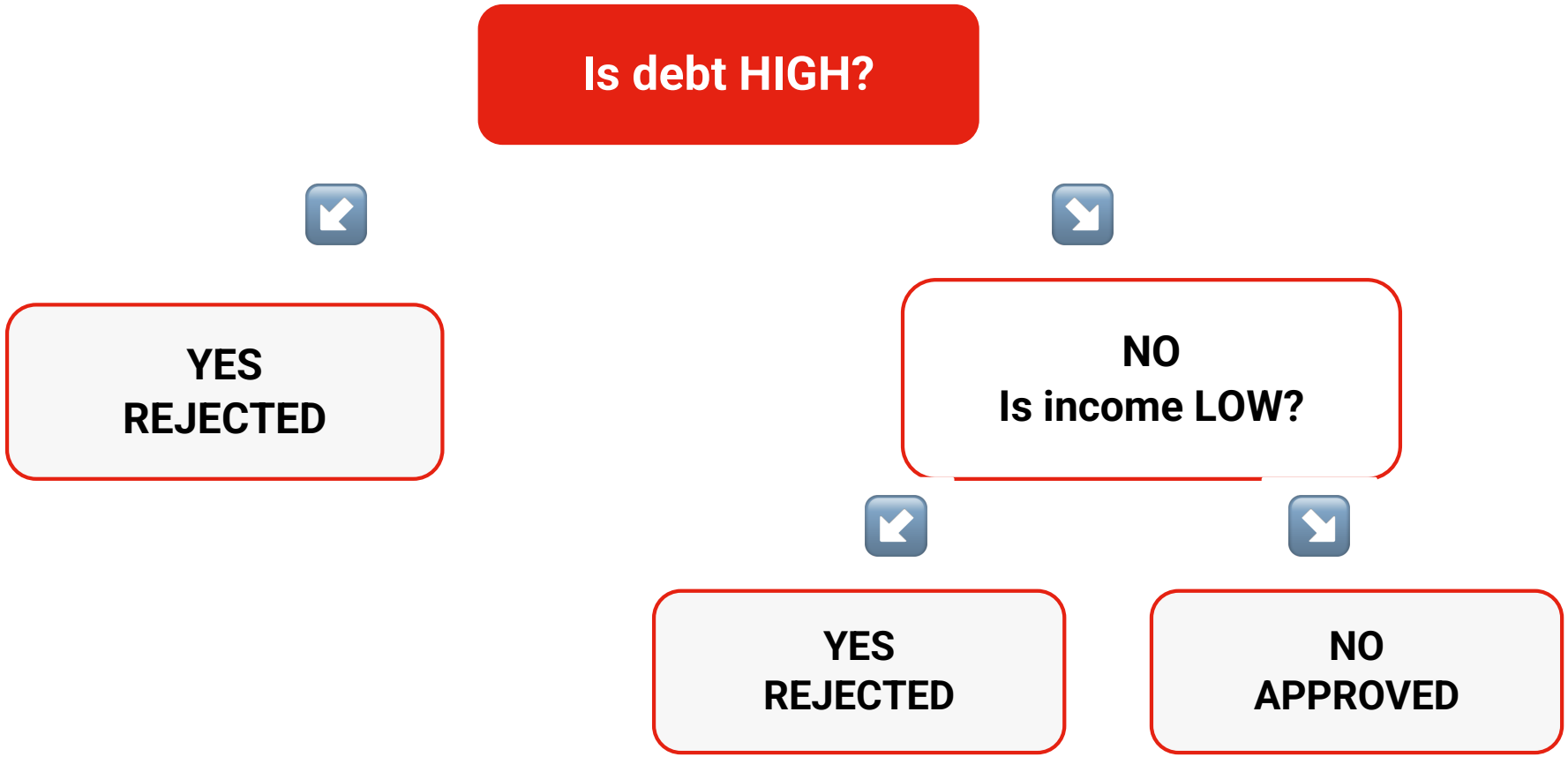
Model rule:
Approve only when
Debt = Low AND
Income ≠ Low

Training accuracy
5/5 = 100%
Training error = 0%

Model fits well on training data

LA TROBE UNIVERSITY
All kinds of clever

A Simple Decision Tree Model representation – Model on tree



A decision tree tests one attribute at a time until it reaches a class.

LA TROBE UNIVERSITY
All kinds of clever

Lets check the model created on Ten Unseen Testing Samples

Apply the same boundary without changing it. Compare each prediction with the actual class.

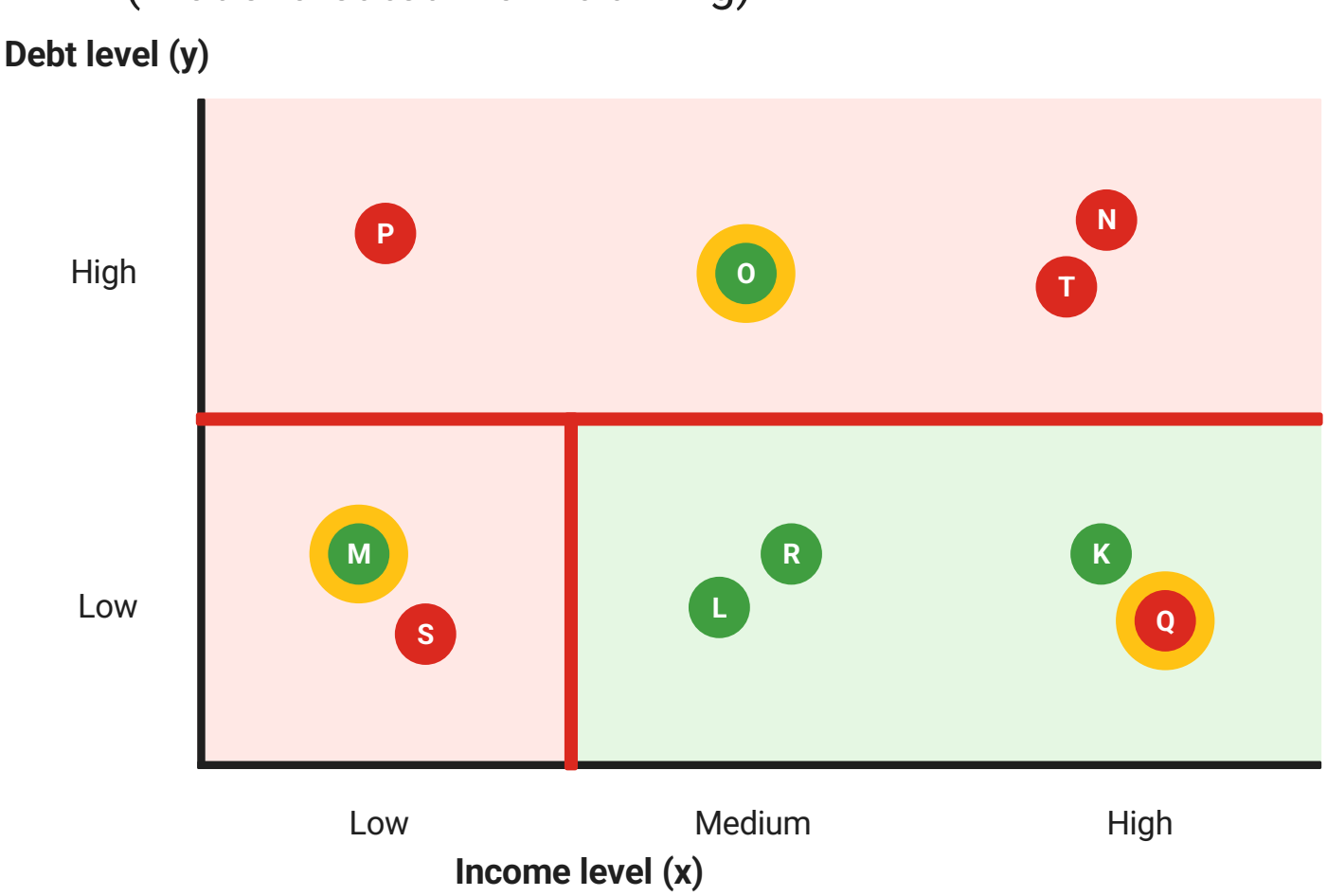
ID	Income	Debt	Actual	Predicted	Result
K	High	Low	Approved	Approved	Correct
L	Medium	Low	Approved	Approved	Correct
M	Low	Low	Approved	Rejected	Incorrect
N	High	High	Rejected	Rejected	Correct
O	Medium	High	Approved	Rejected	Incorrect
P	Low	High	Rejected	Rejected	Correct
Q	High	Low	Rejected	Approved	Incorrect
R	Medium	Low	Approved	Approved	Correct
S	Low	Low	Rejected	Rejected	Correct
T	High	High	Rejected	Rejected	Correct

Result: 7/10 correct. Three unseen applicants are misclassified.

LA TROBE UNIVERSITY
All kinds of clever

Testing the Same Boundary

Ten unseen applicants are classified using exactly the same fixed rule (Model created from training)



Approved

Rejected

Model rule:
Approve only when
Debt = Low AND
Income ≠ Low

Observed test result
7/10 correct = 70%
Test error = 3/10 = 30%
(yellow rings)

How can we build a model, so that it has a good generalization?

It estimates the expected generalisation error on similar future applicants. We can see significant difference between performance of training and testing data.

LA TROBE UNIVERSITY
All kinds of clever

From a Simple Rule to a Larger Dataset

WHY MOVE ON?

Our five training samples produced a simple rule with 100% training accuracy. But a perfect training score does not prove that the model will generalise.

NEXT EXPERIMENT

Use a larger two-class dataset with numeric features x1 and x2. Build decision trees with increasing numbers of nodes.

TWO CLASS PROBLEM

5400 instances

- 5000 instances generated from a Gaussian centered at (10,10)
- 400 noisy instances added

o : 5400 instances

- Generated from a uniform distribution

WHAT TO OBSERVE

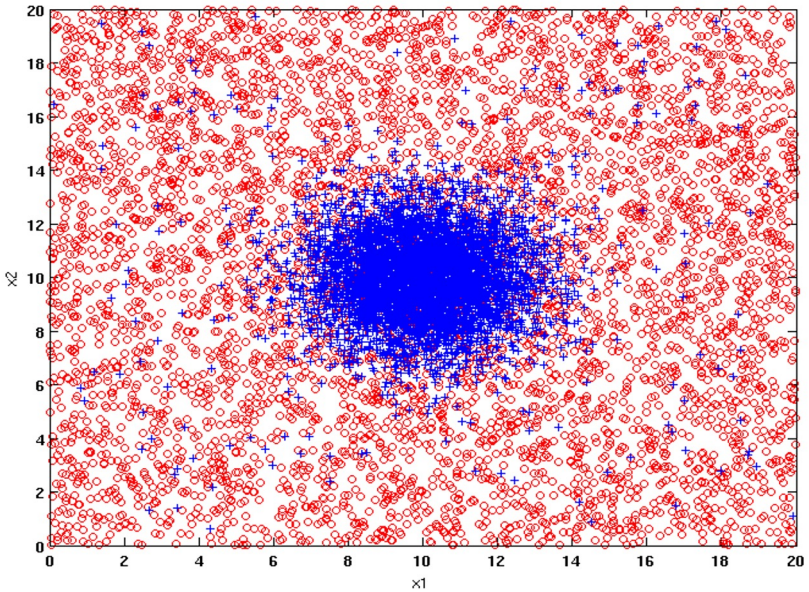
1. How each new split adds a decision boundary.
2. How training error changes as the tree grows.
3. Whether a closer training fit improves—or harms—performance on unseen data.

SETUP

10% training and 90% unseen testing are used for this demonstration only.

GUIDING QUESTION

Which tree is complex enough to learn the pattern, but simple enough to generalise

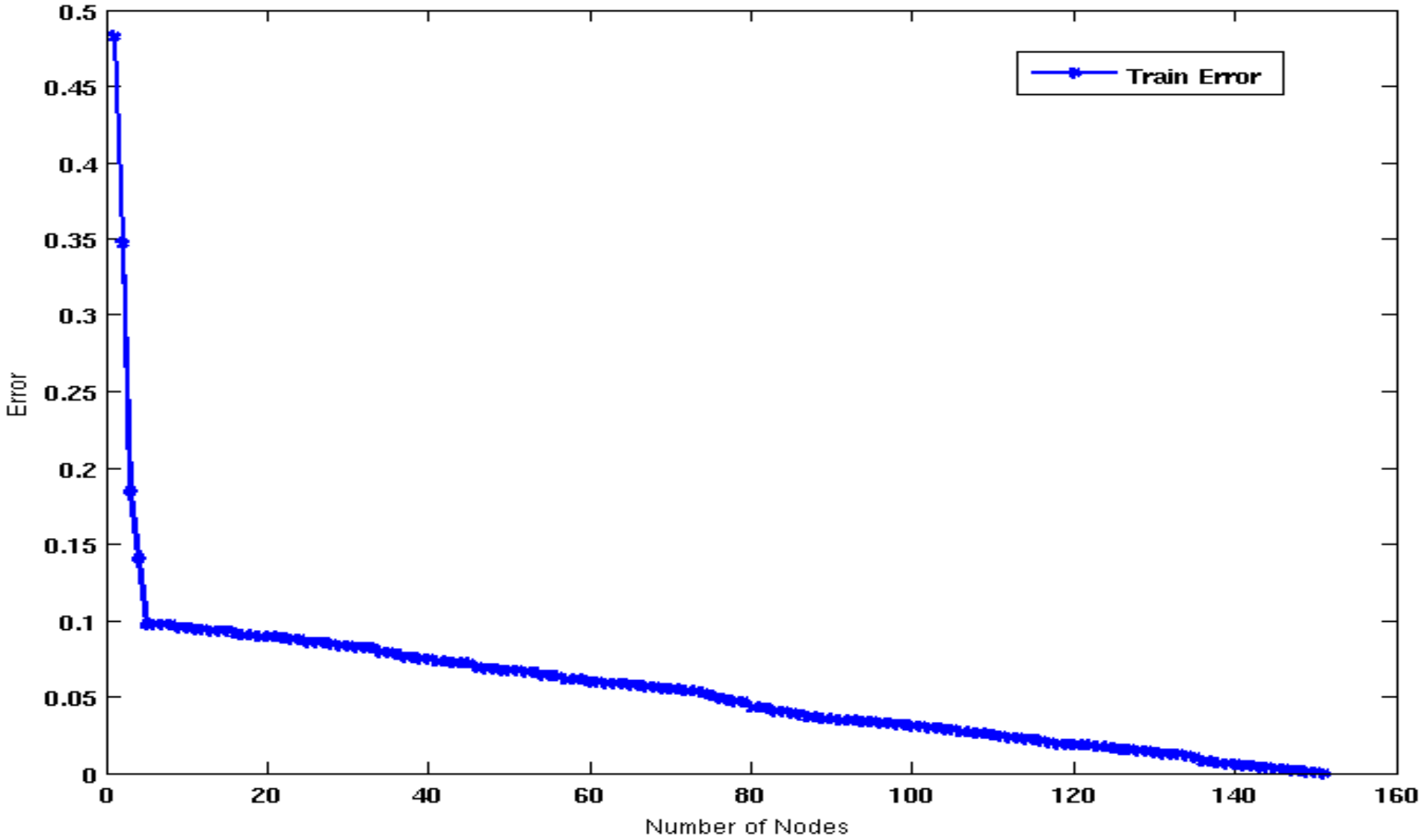


?

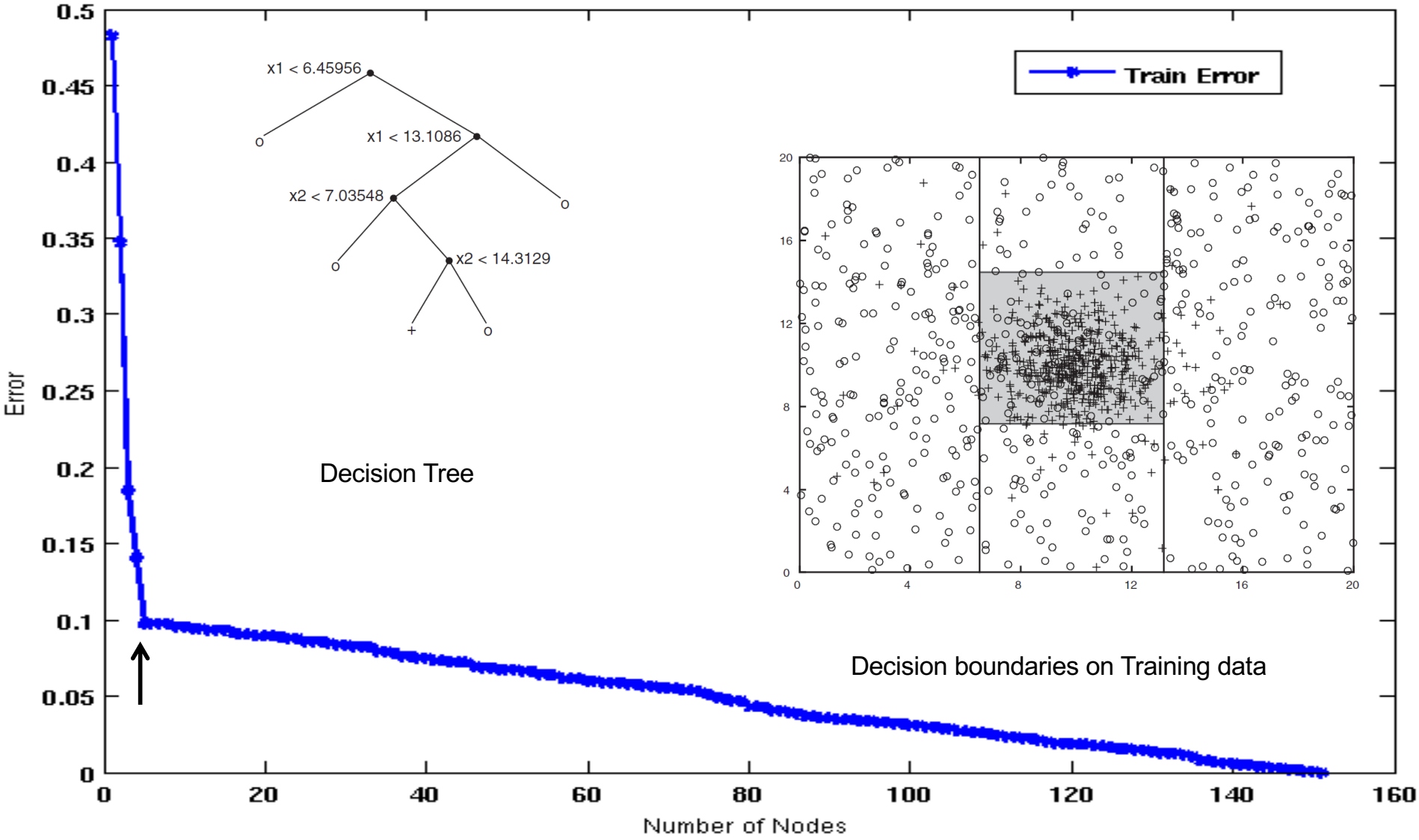
LA TROBE UNIVERSITY
All kinds of clever

Observe Train Error

Increasing number of nodes in Decision Trees

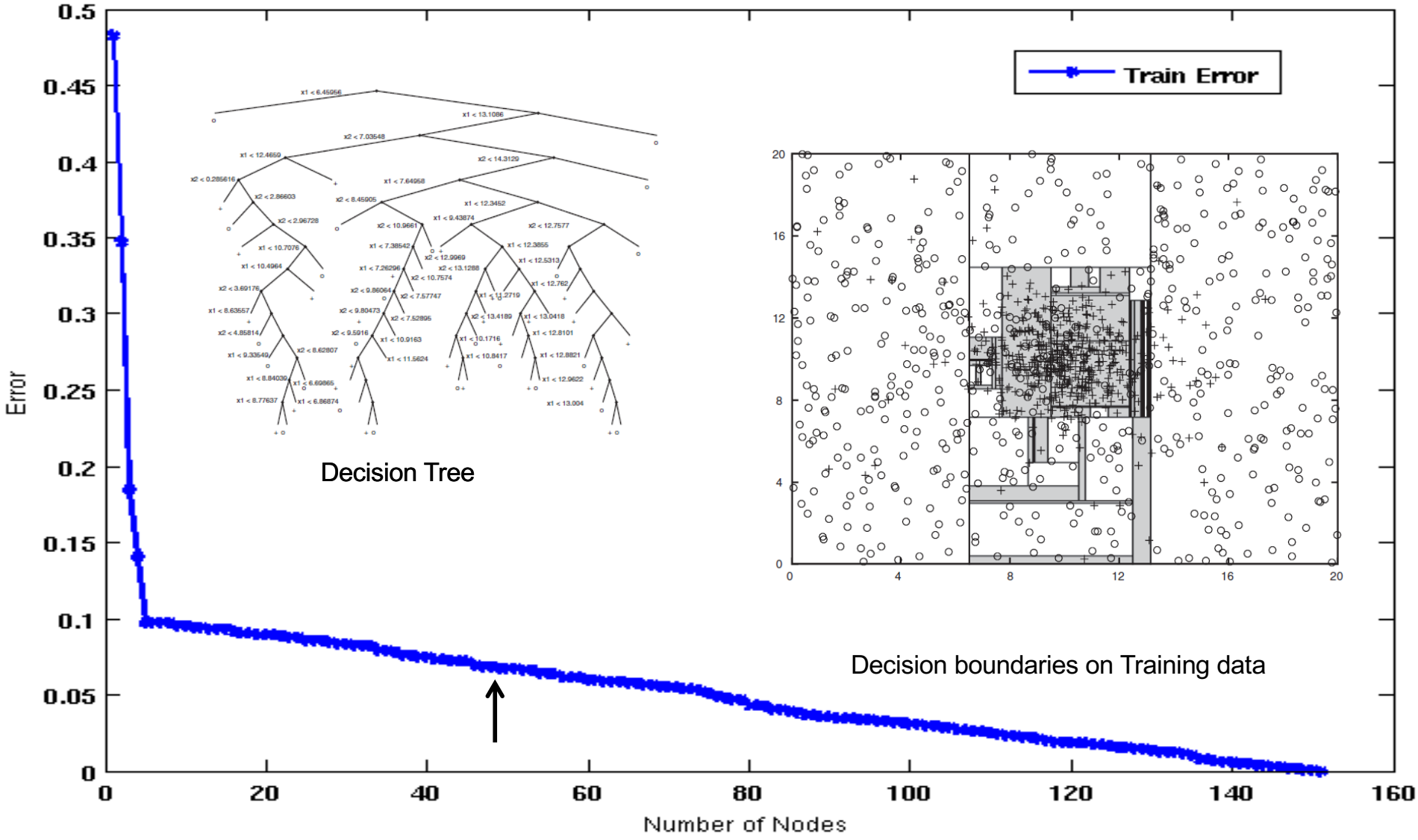


Decision Tree with 4 nodes



LA TROBE UNIVERSITY
All kinds of clever

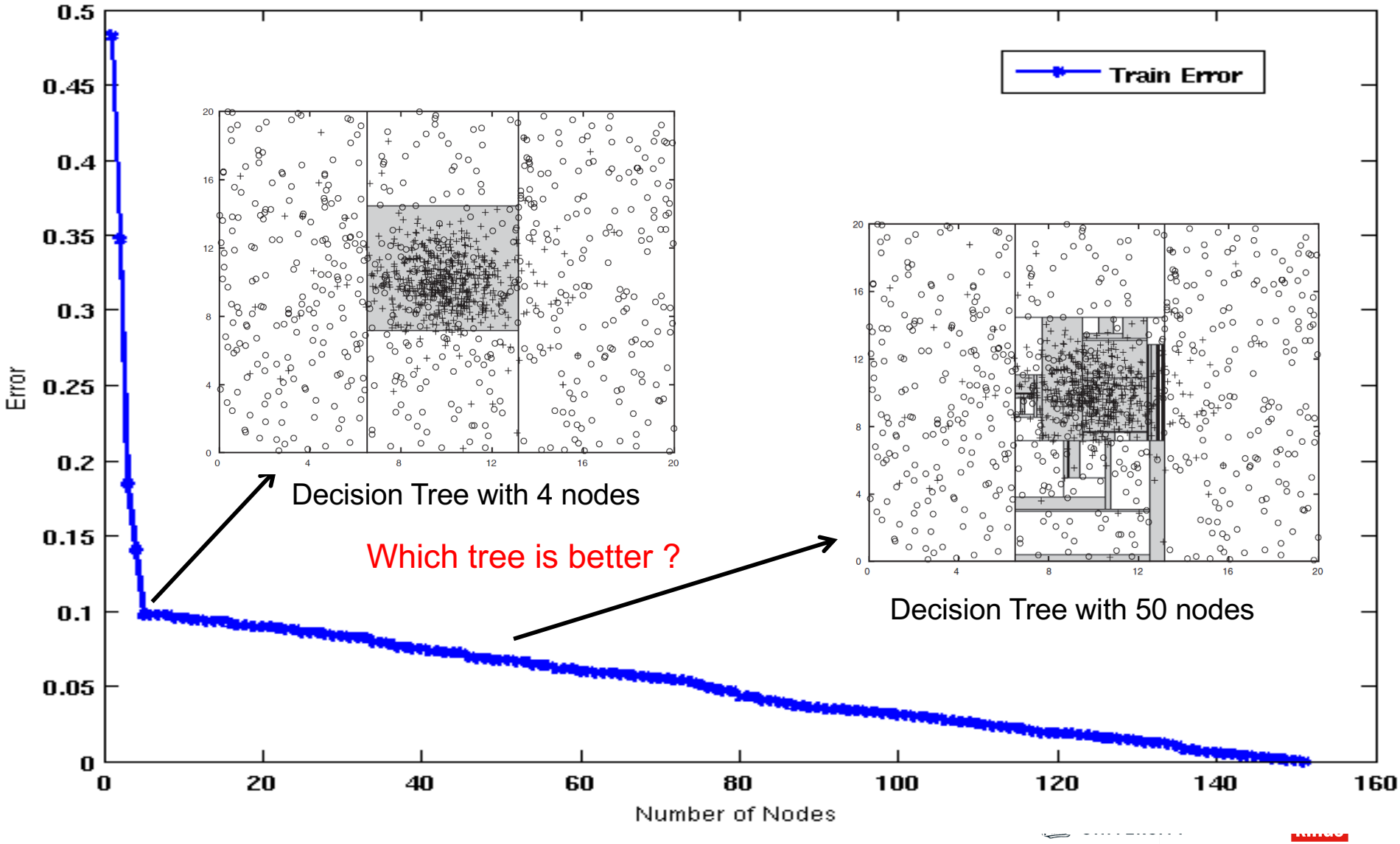
Decision Tree with 50 nodes



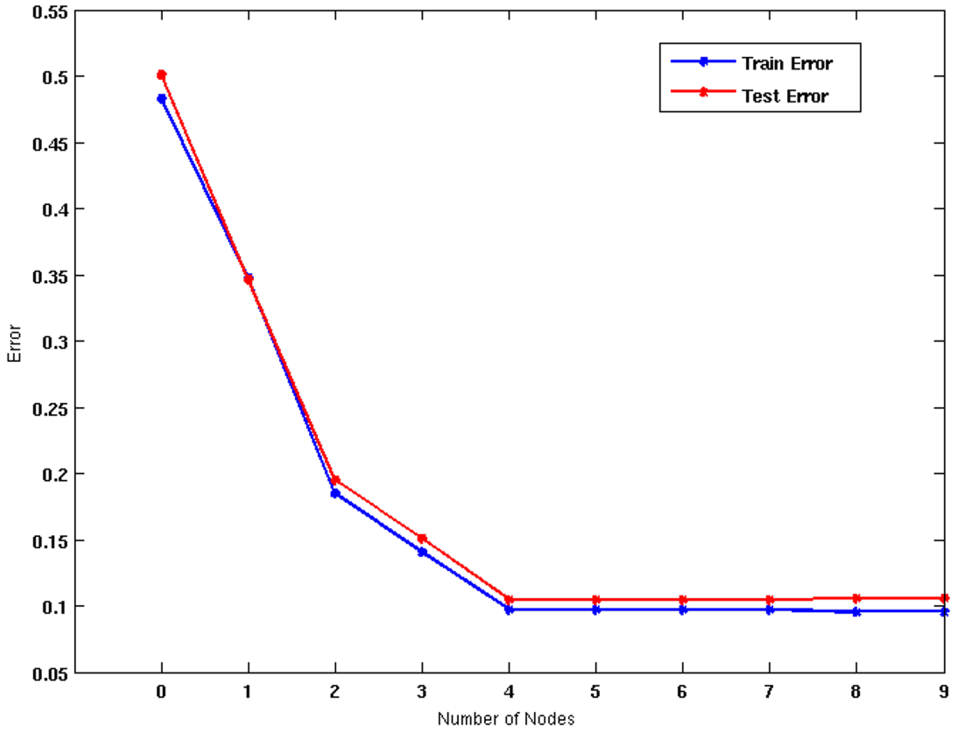
Decision Tree

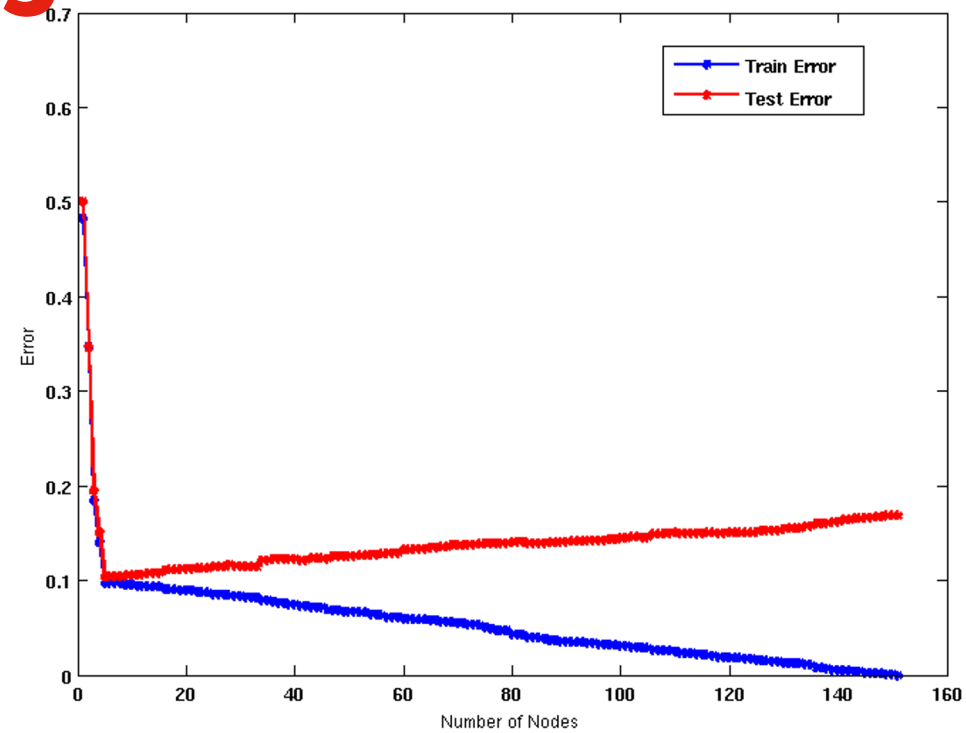
Decision boundaries on Training data

Which tree is better?



Model Overfitting





•As the model becomes more and more complex, test errors can start increasing even though training error may be decreasing

Underfitting: when model is too simple, both training and test errors are large

Overfitting: when model is too complex, training error is small but test error is large

LA TROBE UNIVERSITY
All kinds of clever

Why Models Overfit

Overfitting occurs when a model **learns details and noise from** the training sample instead of the underlying pattern.

Too little data
A few examples make isolated or noisy cases appear more important than they really are.

Too much complexity
The tree keeps splitting until it memorises exceptions in the training set.

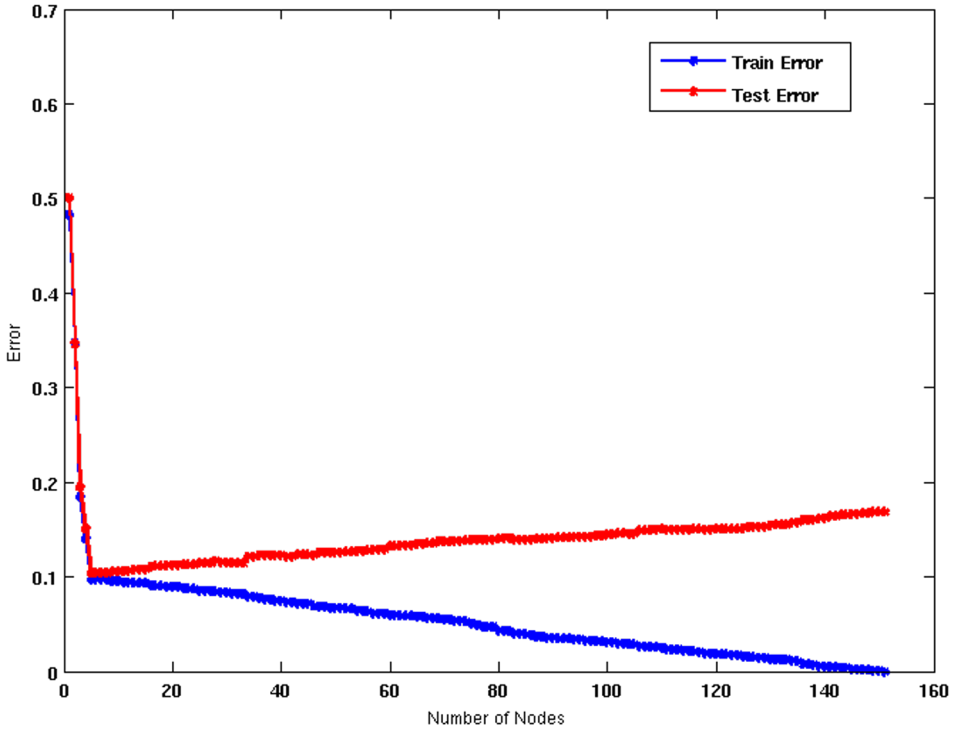
Unrepresentative data
If training data differs from future data, the learned rules will not transfer reliably.

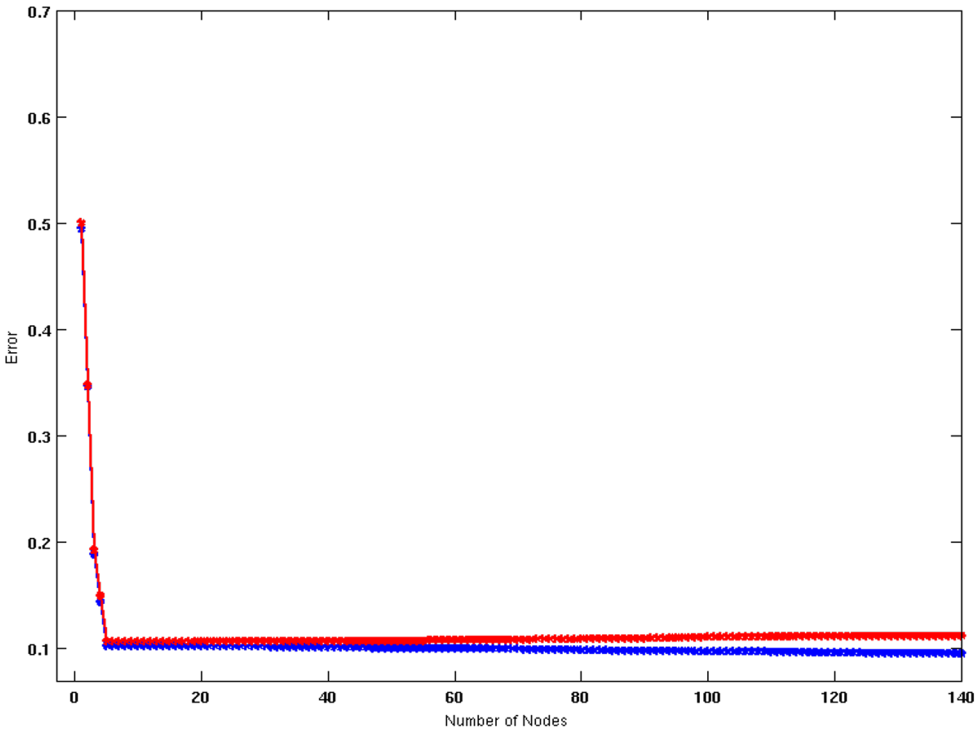
SIGNAL OF OVERFITTING
Very low training error + noticeably higher test error

Question: Which remedy would help—more representative data, a simpler tree, or both?

LA TROBE UNIVERSITY
All kinds of clever

Model Overfitting – Impact of Training Data Size





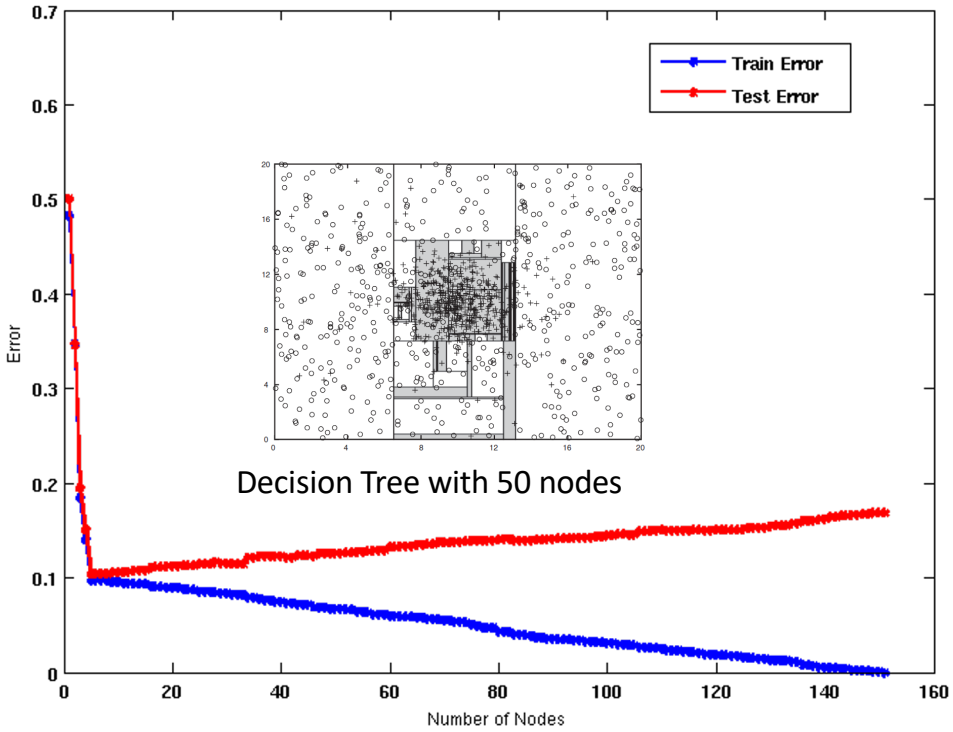
Using twice the number of data instances

- Increasing the size of training data reduces the difference between training and testing errors at a given size of model

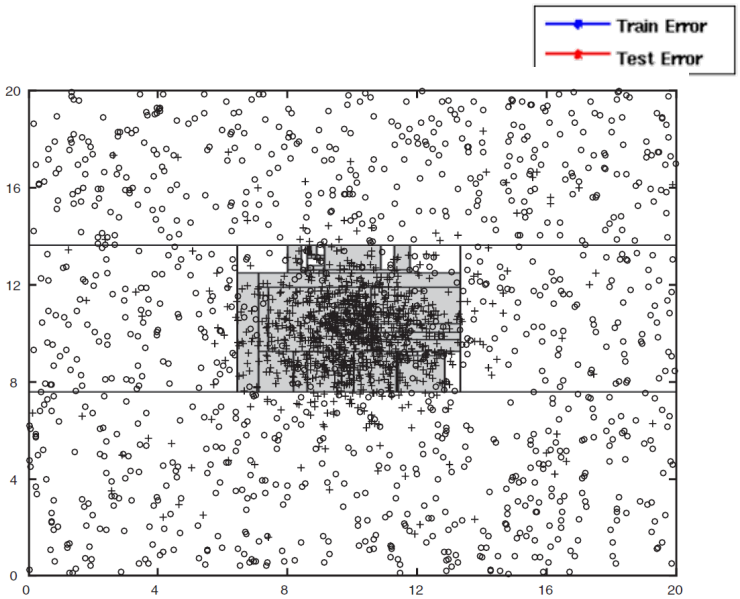
LA TROBE UNIVERSITY
All kinds of clever

Model Overfitting – Impact of Training Data Size

More Data Stabilises the Tree



Decision Tree with 50 nodes



Decision Tree with 50 nodes

Using twice the number of data instances

- Increasing the size of training data reduces the difference between training and testing errors at a given size of model
- A 50-node tree trained on **more examples** is less likely to build rules around isolated noise.

LA TROBE UNIVERSITY
All kinds of clever

Notes on Overfitting

- Overfitting results in decision trees that are more complex than necessary (left figure)
- Training error does not provide a good estimate of how well the tree will perform on previously unseen records. It underestimates future error.
- Need ways for estimating generalization errors

LA TROBE UNIVERSITY
All kinds of clever

Model Selection

After building several decision trees, ***how do we decide which one to use?***
Our goal is to select a model that ***generalises well*** *rather than one that simply fits the training data.*

1. Performed during model building
2. Purpose is to ensure that model is not overly complex (to avoid overfitting)
3. Need to estimate generalization error

   Estimate how each tree may perform on unseen data using:

   a. validation data, or

   b. training error plus a complexity penalty.
4. Select the balanced tree

   a. Choose the tree with the lowest estimated generalisation error

LA TROBE UNIVERSITY
All kinds of clever

Model Selection:

Divide training data into two parts:

- Training set:
  - use for model building
- Validation set:
  - Note: validation set is not the same as test set

Drawback:

- Less data available for training

LA TROBE UNIVERSITY
All kinds of clever

Model Selection:
Estimating the Complexity of Decision Trees

Pessimistic Error Estimate of decision tree T with k leaf nodes:

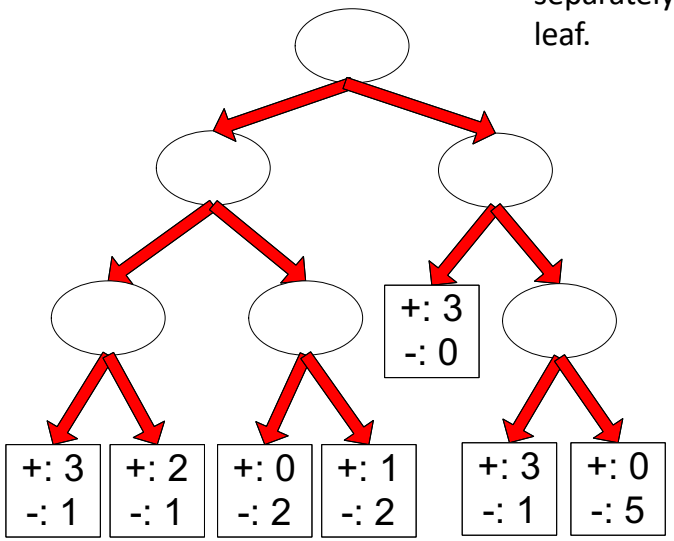
$$err_{gen}(T) = err(T) + \Omega \frac{k}{N_{train}}$$

- err(T): error rate on all training records
- Ω: trade-off hyper-parameter (complexity penalty)
  - Relative cost of adding a leaf node (Controls how strongly additional leaves are penalised.)
- k: number of leaf nodes
- N_{train}: total number of training records

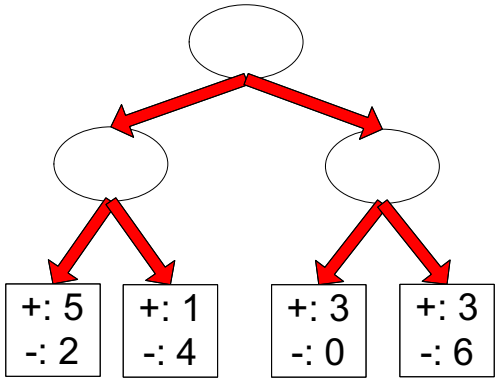
LA TROBE UNIVERSITY
All kinds of clever

Estimating the Complexity of Decision Trees: Example

It assigns a predicted class separately at each final leaf.



Decision Tree, T_L



Decision Tree, T_R

If each leaf node is labeled according to the majority class of training instances that reach the node,

$e(T_L) = 4/24$

$e(T_R) = 6/24$

Based on training error rates alone, Tree L is preferred over Tree R, though L is more complex than R

Left tree T_L

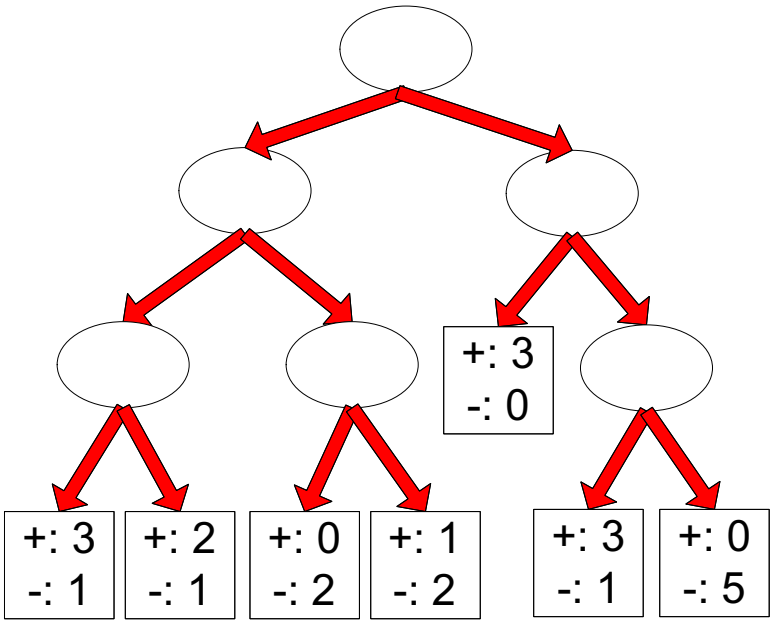
Leaf counts	Predicted class	Errors
$+3, -1$	$+$	1
$+2, -1$	$+$	1
$+0, -2$	$-$	0
$+1, -2$	$-$	1
$+3, -0$	$+$	0
$+3, -1$	$+$	1
$+0, -5$	$-$	0
Total		**4**

Right tree T_R

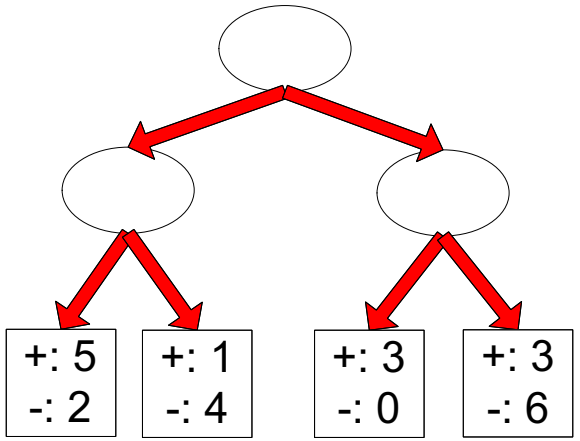
Leaf counts	Predicted class	Errors
$+5, -2$	$+$	2
$+1, -4$	$-$	1
$+3, -0$	$+$	0
$+3, -6$	$-$	3
Total		**6**

LA TROBE UNIVERSITY
All kinds of clever

Estimating the Complexity of Decision Trees: Example



Decision Tree, T_L



Decision Tree, T_R

k is the number of leaf for each tree
kL = 7
kR = 4

$$err_{gen}(T) = err(T) + \Omega \times \frac{k}{N_{train}}$$

Assume Ω = 1

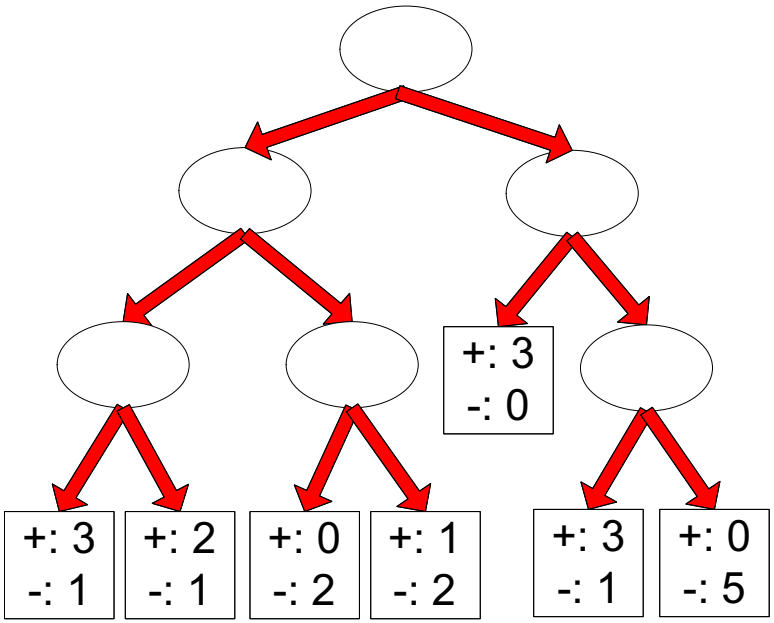
Since Tree R has a lower generalization error rate, we prefer Tree R over Tree L.

$e_{gen}(T_L)$ = 4/24 + 1*7/24 = 11/24 = 0.458

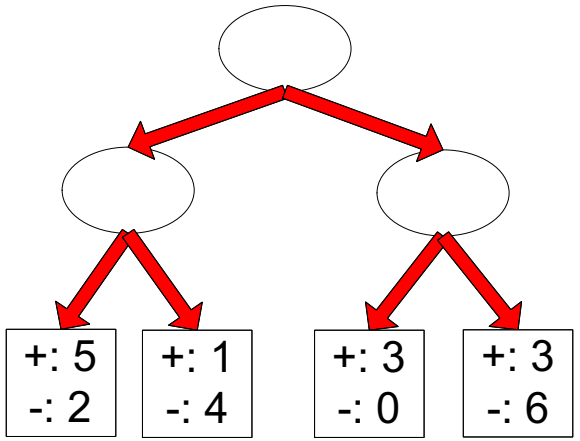
$e_{gen}(T_R)$ = 6/24 + 1*4/24 = 10/24 = 0.417

LA TROBE UNIVERSITY
All kinds of clever

Estimating the Complexity of Decision Trees: Example



Decision Tree, T_L



Decision Tree, T_R

$$err_{gen}(T) = err(T) + \Omega \times \frac{k}{N_{train}}$$

Assume Ω = 0.5

Different choices of Ω can change our preference
(this case implies that expanding a node is less costly than misclassifying a training instance)

$e_{gen}(T_L) = 4/24 + 0.5*7/24 = 7.5/24$

$e_{gen}(T_R) = 6/24 + 0.5*4/24 = 8/24$

LA TROBE UNIVERSITY
All kinds of clever

Estimating the Complexity of Decision Trees: Example

This example illustrates that different choices of Ω can change our preference of decision trees based on their generalization error estimates.

However, for a given choice of Ω, the error estimate provides an approach for modelling the generalization performance on unseen test instances.

The value of Ω can be selected with the help of a validation set.

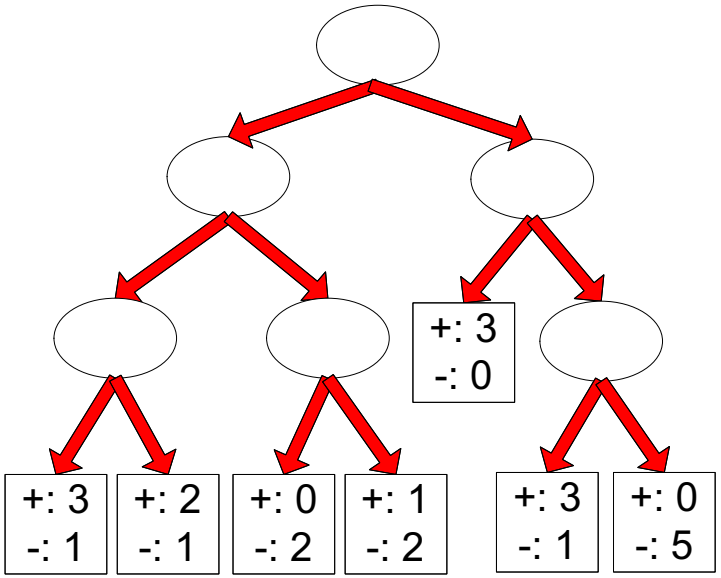
The approach for estimating generalization error rate is also termed as the pessimistic error estimate. It is called pessimistic as it assumes the generalization error rate to be worse than the training error rate, especially if the model is very complex or trained on limited data

On the other hand, simply using the training error rate as an estimate of the generalization error rate is called the **optimistic error estimate** or the **resubstitution estimate**

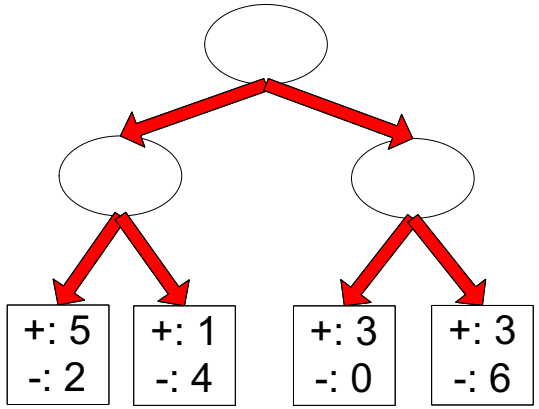
LA TROBE UNIVERSITY
All kinds of clever

Estimating the Complexity of Decision Trees

- Resubstitution Estimate:
  - Using training error as an optimistic estimate of generalization error
  - Referred to as optimistic error estimate



Decision Tree, T_L



Decision Tree, T_R

$e(T_L) = 4/24$

$e(T_R) = 6/24$

LA TROBE UNIVERSITY
All kinds of clever

How to address overfitting

Pre-Pruning (Early Stopping Rule)

- Stop the algorithm before it becomes a fully-grown tree
- Typical stopping conditions for a node:
  - Stop if all instances belong to the same class
  - Stop if all the attribute values are the same
- More restrictive conditions:
  - Stop if number of instances is less than some user-specified threshold
  - Stop if class distribution of instances are independent of the available features (e.g., using χ^2 test)
  - Stop if expanding the current node does not improve impurity measures (e.g., Gini or information gain).

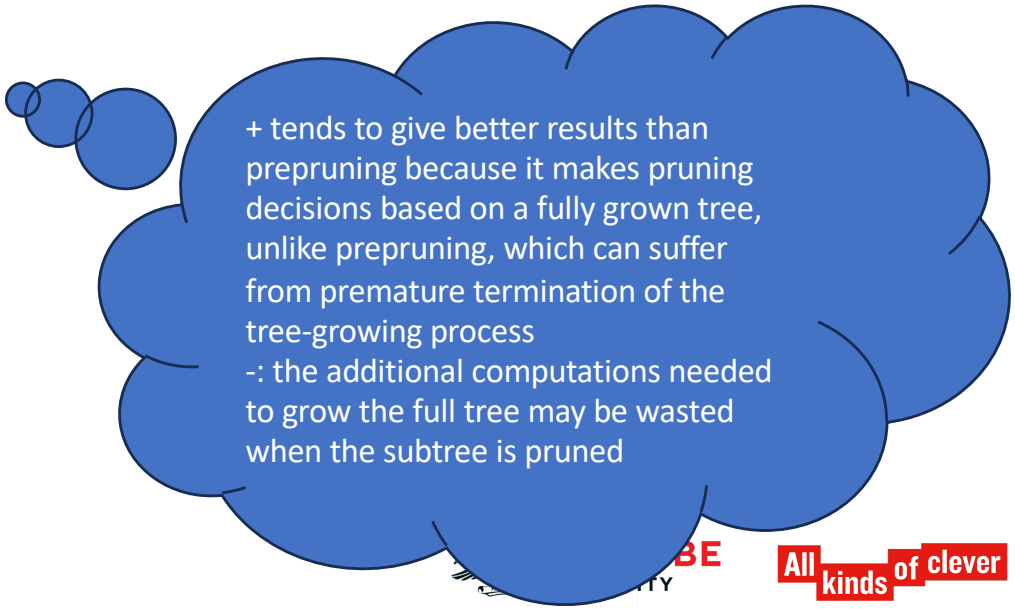
+ avoids the computations associated with generating overly complex subtrees that overfit the training data.
-: even if no significant gain is obtained using one of the existing splitting criterion, subsequent splitting may result in better subtrees

LA TROBE UNIVERSITY
All kinds of clever

How to address overfitting

Post-pruning

- Grow decision tree to its entirety
- Subtree replacement
  - Trim the nodes of the decision tree in a bottom-up fashion
  - If generalization error improves after trimming, replace sub-tree by a leaf node
  - Class label of leaf node is determined from majority class of instances in the sub-tree



All kinds of clever

Example of Post-Pruning

Class = Yes	20
Class = No	10
Error = 10/30	

Training Error (Before splitting) = 10/30

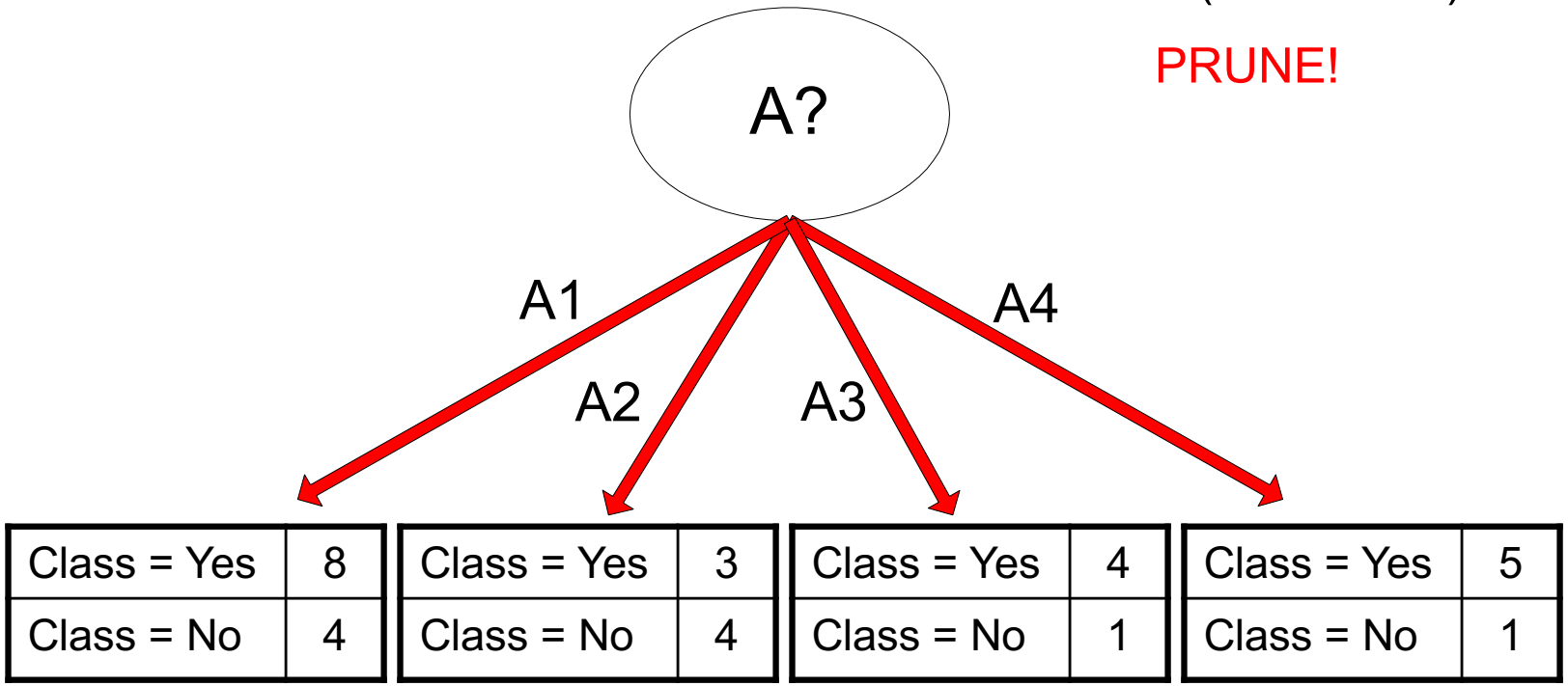
Pessimistic error = (10 + 0.5)/30 = 10.5/30

Training Error (After splitting) = 9/30

Pessimistic error (After splitting)

= (9 + 4 × 0.5)/30 = 11/30

PRUNE!



LA TROBE UNIVERSITY
All kinds of clever

Model evaluation

Purpose:

- To estimate performance of classifier on previously unseen data (test set)

Holdout

- Reserve k% for training and (100-k)% for testing
- Random subsampling: repeated holdout

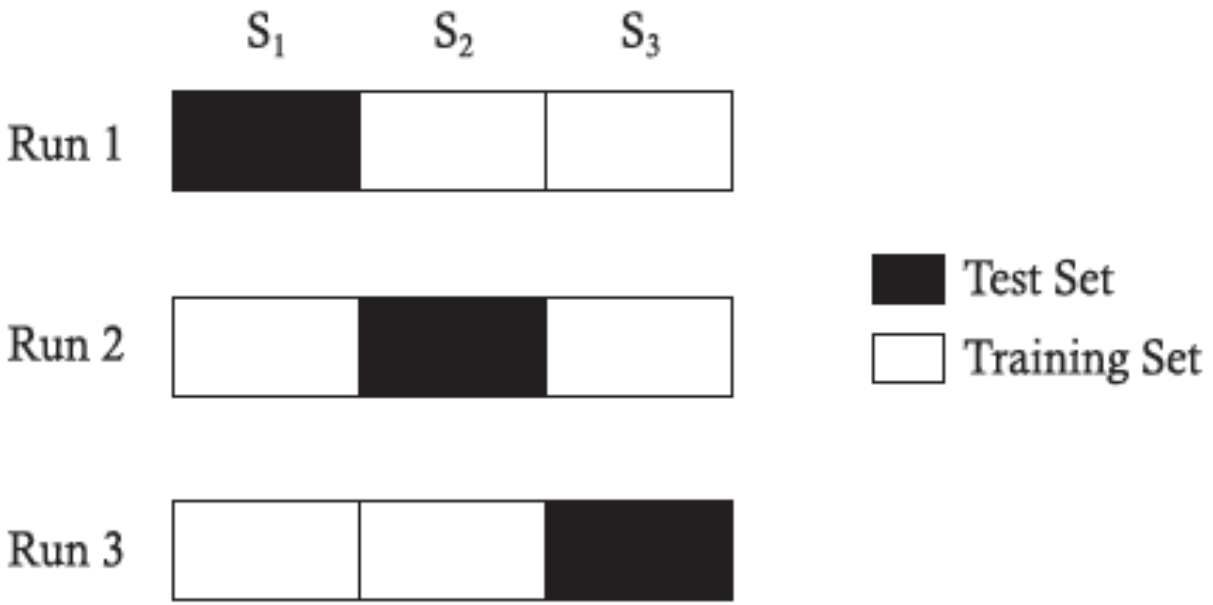
Cross validation

- Partition data into k disjoint subsets
- k-fold: train on k-1 partitions, test on the remaining one
- Leave-one-out: k=n

LA TROBE UNIVERSITY
All kinds of clever

Cross-validation Example

- 3-fold cross-validation



LA TROBE UNIVERSITY
All kinds of clever

Variations on Cross-validation

- Repeated cross-validation
  - Perform cross-validation a number of times
  - Gives an estimate of the variance of the generalization error
- Stratified cross-validation
  - Guarantee the same percentage of class labels in training and test
  - Important when classes are imbalanced and the sample is small
- Others: Use nested cross-validation approach for model selection and evaluation

LA TROBE UNIVERSITY
All kinds of clever

Model evaluation

Metrics for Performance Evaluation

- How to evaluate the performance of a model?

Methods for Performance Evaluation

- How to obtain reliable estimates?

Methods for Model Comparison

- How to compare the relative performance among competing models?

LA TROBE UNIVERSITY
All kinds of clever

Metrics for performance evaluation

Focus on the predictive capability of a model

- Rather than how fast it takes to classify or build models, scalability, etc

Confusion Matrix:

		PREDICTED CLASS	
		Class=Yes	Class=No
ACTUAL CLASS	Class= Yes	A	B
	Class= No	C	D

A: TP (true positive)

B: FN (false negative)

C: FP (false positive)

D: TN (true negative)

LA TROBE UNIVERSITY
All kinds of clever

CONFUSION MATRIX

		PREDICTED CLASS	
		Class=Yes *positive*	Class=No *negative*
ACTUAL CLASS	Class= Yes	**A** *Ture positive*	**B** *False negative*
	Class= No	**C** *False positive*	**D** *Ture negative*

Note: In *true/false positive/negative*, *true* or *false* refers to the assigned classification being correct or incorrect, while *positive* or *negative* refers to assignment to the positive or the negative category.

- **Classification error**
  - Type 1 (reject a true positive)
    - e.g. Healthy people incorrectly identified as sick
    - *What is true positive? actual positives that are correctly identified as such*
  - Type 2 (fail to reject a false negative)
    - e.g. Sick people incorrectly identified as healthy

LA TROBE UNIVERSITY
All kinds of clever

Example 1: Confusion Matrix Calculation

		PREDICTED CLASS	
		Class=Yes *positive*	Class=No *negative*
ACTUAL CLASS	Class= Yes	**A** *Ture positive*	**B** *False negative*
	Class= No	**C** *False positive*	**D** *Ture negative*

Sample ID	Actual Class	Predicted class
#1	1	1
#2	2	2
#3	2	2
#4	1	2
#5	2	1
#6	1	1
#7	1	1
#8	2	1

Confusion Matrix

		Predicted class	
Actual Class		1	2
	1	**3**	**1**
	2	2	2

LA TROBE UNIVERSITY
All kinds of clever

Metrics for performance evaluation

		PREDICTED CLASS	
		Class=Yes	Class=No
ACTUAL CLASS	Class= Yes	A	B
	Class= No	C	D

A: TP (true positive)

B: FN (false negative)

C: FP (false positive)

D: TN (true negative)

Most widely-used metric:

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

TROBE ERSITY All kinds of clever

Limitation of accuracy

Consider a 2-class problem

- Number of Class 0 examples = 9990
- Number of Class 1 examples = 10

If model predicts everything to be class 0, accuracy is 9990/10000 = 99.9 %

- Accuracy is misleading because model does not detect any class 1 example

LA TROBE UNIVERSITY
All kinds of clever

Cost Matrix

	PREDICTED CLASS		
ACTUAL CLASS	**C(i | j)**	**Class=Yes**	**Class=No**
	Class=Yes	C(Yes|Yes)	**C(No|Yes)**
	Class=No	**C(Yes|No)**	C(No|No)

C(i | j): Cost of misclassifying class **j** example as class **i**

LA TROBE UNIVERSITY
All kinds of clever

Computing cost of classification

Cost Matrix	PREDICTED CLASS		
	C(i|j)	+	-
ACTUAL CLASS	+	-1	100
	-	1	0

Model M_1	PREDICTED CLASS		
		+	-
ACTUAL CLASS	+	150	40
	-	60	250

Accuracy = 80%

Cost = 150 * (-1) + 40 * 100
+ 60 * 1 + 250 * 0 = 3910

Model M_2	PREDICTED CLASS		
		+	-
ACTUAL CLASS	+	250	45
	-	5	200

Accuracy = 90%

Cost = 250 * (-1) + 45 * 100
+ 5 * 1 + 200 * 0 = 4255

LA TROBE UNIVERSITY
All kinds of clever

Cost-sensitive measures

		PREDICTED CLASS	
		Class =Yes	Class =No
ACTUAL CLASS	Class= Yes	**a** (TP)	b (FN)
	Class= No	c (FP)	**d** (TN)

$$\text{Precision (p)} = \frac{a}{a+c}$$

$$\text{Recall (r)} = \frac{a}{a+b}$$

$$\text{F - measure (F)} = \frac{2rp}{r+p} = \frac{2a}{2a+b+c}$$

- Precision is biased towards C(Yes|Yes) & C(Yes|No)
- Recall is biased towards C(Yes|Yes) & C(No|Yes)
- F-measure is biased towards all except C(No|No)

LA TROBE UNIVERSITY
All kinds of clever

Model evaluation

Metrics for Performance Evaluation

- How to evaluate the performance of a model?

Methods for Performance Evaluation

- How to obtain reliable estimates?

Methods for Model Comparison

- How to compare the relative performance among competing models?

LA TROBE UNIVERSITY
All kinds of clever

Methods for Performance Evaluation

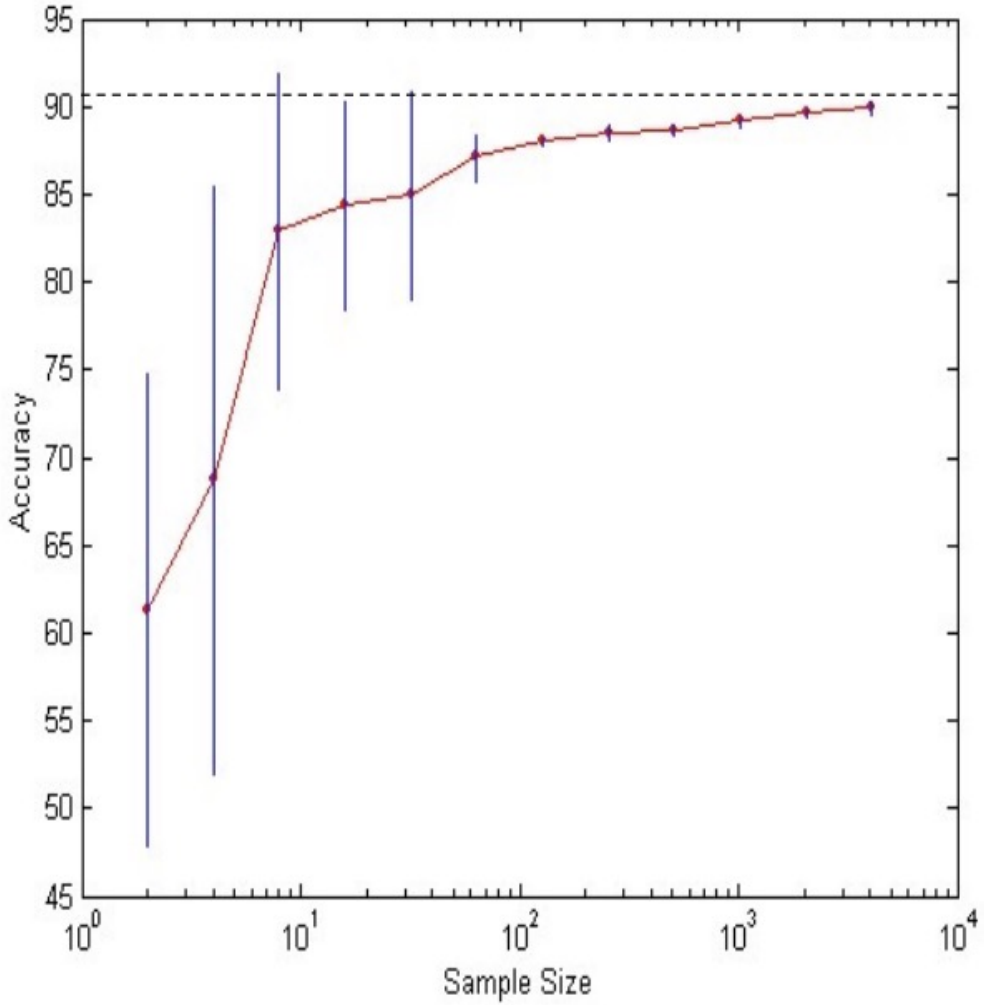
How to obtain a reliable estimate of performance?

Performance of a model may depend on other factors besides the learning algorithm:

- Class distribution
- Cost of misclassification
- Size of training and test sets

LA TROBE UNIVERSITY
All kinds of clever

Learning Curve



- Learning curve shows how accuracy changes with varying sample size
- Requires a sampling schedule for creating learning curve
- Effect of small sample size:
  - Bias in the estimate
  - Variance of estimate

LA TROBE UNIVERSITY
All kinds of clever

Model evaluation

Metrics for Performance Evaluation

- How to evaluate the performance of a model?

Methods for Performance Evaluation

- How to obtain reliable estimates?

Methods for Model Comparison

- How to compare the relative performance among competing models? (Next Week)

LA TROBE UNIVERSITY
All kinds of clever

Summary

Overfitting

- Reasons
- How to avoid overfitting
- Model selection
    - Generalization error estimation
    - Validation dataset
    - Pessimistic error estimation
- Prepruning and postpruning

Model evaluation

- Training, validation, test datasets
- Confusion matrix, Accuracy, etc
- Learning curve
- Next Week: Receiver Operating Characteristic

LA TROBE UNIVERSITY
All kinds of clever



Thank you

latrobe.edu.au

La Trobe University
CRICOS Provider Code Number 00115M

© Copyright La Trobe University 2018